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Java SE 27 & JDK 27

Class AtomicIntegerFieldUpdater<T>

java.lang.Object
    java.util.concurrent.atomic.AtomicIntegerFieldUpdater<T>

Type Parameters:

T - The type of the object holding the updatable field

```
public abstract class AtomicIntegerFieldUpdater<T>
extends Object
```

A reflection-based utility that enables atomic updates to designated non-static `volatile int` fields of designated classes, providing a subset of the functionality of class `VarHandle` that should be used instead. This class is designed for use in atomic data structures in which several fields of the same node are independently subject to atomic updates.

Note that the guarantees of the `compareAndSet` method in this class are weaker than in other atomic classes. Because this class cannot ensure that all uses of the field are appropriate for purposes of atomic access, it can guarantee atomicity only with respect to other invocations of `compareAndSet` and `set` on the same updater.

Object arguments for parameters of type T that are not instances of the class passed to `newUpdater(Class, String)` will result in a `ClassCastException` being thrown.

Since:

1.5

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	AtomicIntegerFieldUpdate	Protected do-nothing constructor for use by subclasses.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
final int		Atomically updates (with memory effects as

	accumulateAndGet(T obj, int x, IntBinaryOperator accumu	specified by VarHandle.compareAndSet(Object...)) the field of the given object managed by this updater with the results of applying the given function to the current and given values, returning the updated value.
int	**addAndGet**(T obj, int delta)	Atomically adds the given value to the current value of the field of the given object managed by this updater.
abstract boolean	**compareAndSet**(T obj, int expect, int update)	Atomically sets the field of the given object managed by this updater to the given updated value if the current value == the expected value.
int	**decrementAndGet**(T obj)	Atomically decrements by one the current value of the field of the given object managed by this updater.
abstract int	**get**(T obj)	Returns the current value held in the field of the given object managed by this updater.
final int	**getAndAccumulate**(T obj, int x, IntBinaryOperator accumu	Atomically updates (with memory effects as specified by VarHandle.compareAndSet(Object...)) the field of the given

		object managed by this updater with the results of applying the given function to the current and given values, returning the previous value.
int	**getAndAdd**(T obj, int delta)	Atomically adds the given value to the current value of the field of the given object managed by this updater.
int	**getAndDecrement**(T obj)	Atomically decrements by one the current value of the field of the given object managed by this updater.
int	**getAndIncrement**(T obj)	Atomically increments by one the current value of the field of the given object managed by this updater.
int	**getAndSet**(T obj, int newValue)	Atomically sets the field of the given object managed by this updater to the given value and returns the old value.
final int	**getAndUpdate**(T obj, IntUnaryOperator updateF	Atomically updates (with memory effects as specified by VarHandle.compareAndSet(Object...)) the field of the given object managed by this updater with the results

		of applying the given function, returning the previous value.
int	**incrementAndGet**(T obj)	Atomically increments by one the current value of the field of the given object managed by this updater.
abstract void	**lazySet**(T obj, int newValue)	Eventually sets the field of the given object managed by this updater to the given updated value.
static <U> AtomicIntegerFie <U>	**newUpdater** (Class<U> tclass, String fieldName)	Creates and returns an updater for objects with the given field.
abstract void	**set**(T obj, int newValue)	Sets the field of the given object managed by this updater to the given updated value.
final int	**updateAndGet**(T obj, IntUnaryOperator updateF	Atomically updates (with memory effects as specified by VarHandle.compareAndSet(Object...)) the field of the given object managed by this updater with the results of applying the given function, returning the updated value.
abstract boolean		Atomically sets the field of the given object managed by this updater

	weakCompareAndSet (T obj, int expect, int update)	to the given updated value if the current value == the expected value.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

AtomicIntegerFieldUpdater

```
protected AtomicIntegerFieldUpdater()
```

Protected do-nothing constructor for use by subclasses.

Method Details

newUpdater

```
public static <U> AtomicIntegerFieldUpdater<U> newUpdater
(Class<U> tclass,
 String fieldName)
```

Creates and returns an updater for objects with the given field. The Class argument is needed to check that reflective types and generic types match.

Type Parameters:

`U` - the type of instances of tclass

Parameters:

`tclass` - the class of the objects holding the field

`fieldName` - the name of the field to be updated

Returns:

the updater

Throws:

`IllegalArgumentException` - if the field is not a volatile integer type

RuntimeException - with a nested reflection-based exception if the class does not hold field or is the wrong type, or the field is inaccessible to the caller according to Java language access control

compareAndSet

```
public abstract boolean compareAndSet(T obj,
                                      int expect,
                                      int update)
```

Atomically sets the field of the given object managed by this updater to the given updated value if the current value == the expected value. This method is guaranteed to be atomic with respect to other calls to `compareAndSet` and `set`, but not necessarily with respect to other changes in the field.

Parameters:

`obj` - An object whose field to conditionally set

`expect` - the expected value

`update` - the new value

Returns:

`true` if successful

weakCompareAndSet

```
public abstract boolean weakCompareAndSet(T obj,
                                          int expect,
                                          int update)
```

Atomically sets the field of the given object managed by this updater to the given updated value if the current value == the expected value. This method is guaranteed to be atomic with respect to other calls to `compareAndSet` and `set`, but not necessarily with respect to other changes in the field.

This operation may fail spuriously and does not provide ordering guarantees, so is only rarely an appropriate alternative to `compareAndSet`.

Parameters:

`obj` - An object whose field to conditionally set

`expect` - the expected value

`update` - the new value

Returns:

`true` if successful

set

```
public abstract void set(T obj,
                         int newValue)
```

Sets the field of the given object managed by this updater to the given updated value. This operation is guaranteed to act as a volatile store with respect to subsequent invocations of `compareAndSet`.

Parameters:

`obj` - An object whose field to set

`newValue` - the new value

lazySet

```
public abstract void lazySet(T obj,
                             int newValue)
```

Eventually sets the field of the given object managed by this updater to the given updated value.

Parameters:

`obj` - An object whose field to set

`newValue` - the new value

Since:

1.6

get

```
public abstract int get(T obj)
```

Returns the current value held in the field of the given object managed by this updater.

Parameters:

`obj` - An object whose field to get

Returns:

the current value

getAndSet

```
public int getAndSet(T obj,
                     int newValue)
```

Atomically sets the field of the given object managed by this updater to the given value and returns the old value.

Parameters:

`obj` - An object whose field to get and set

`newValue` - the new value

Returns:

the previous value

getAndIncrement

```
public int getAndIncrement(T obj)
```

Atomically increments by one the current value of the field of the given object managed by this updater.

Parameters:

`obj` - An object whose field to get and set

Returns:

the previous value

getAndDecrement

```
public int getAndDecrement(T obj)
```

Atomically decrements by one the current value of the field of the given object managed by this updater.

Parameters:

`obj` - An object whose field to get and set

Returns:

the previous value

getAndAdd

```
public int getAndAdd(T obj,
                     int delta)
```

Atomically adds the given value to the current value of the field of the given object managed by this updater.

Parameters:

`obj` - An object whose field to get and set

`delta` - the value to add

Returns:

the previous value

incrementAndGet

```
public int incrementAndGet(T obj)
```

Atomically increments by one the current value of the field of the given object managed by this updater.

Parameters:

`obj` - An object whose field to get and set

Returns:

the updated value

decrementAndGet

```
public int decrementAndGet(T obj)
```

Atomically decrements by one the current value of the field of the given object managed by this updater.

Parameters:

`obj` - An object whose field to get and set

Returns:

the updated value

addAndGet

```
public int addAndGet(T obj,
                     int delta)
```

Atomically adds the given value to the current value of the field of the given object managed by this updater.

Parameters:

`obj` - An object whose field to get and set

`delta` - the value to add

Returns:

the updated value

getAndUpdate

```
public final int getAndUpdate(T obj,
                              IntUnaryOperator updateFunction)
```

Atomically updates (with memory effects as specified by `VarHandle.compareAndSet(Object...)`) the field of the given object managed by this updater with the results of applying the given function, returning the

previous value. The function should be side-effect-free, since it may be re-applied when attempted updates fail due to contention among threads.

Parameters:

`obj` - An object whose field to get and set

`updateFunction` - a side-effect-free function

Returns:

the previous value

Since:

1.8

updateAndGet

```
public final int updateAndGet(T obj,
                              IntUnaryOperator updateFunction)
```

Atomically updates (with memory effects as specified by `VarHandle.compareAndSet(Object...)`) the field of the given object managed by this updater with the results of applying the given function, returning the updated value. The function should be side-effect-free, since it may be re-applied when attempted updates fail due to contention among threads.

Parameters:

`obj` - An object whose field to get and set

`updateFunction` - a side-effect-free function

Returns:

the updated value

Since:

1.8

getAndAccumulate

```
public final int getAndAccumulate
(T obj,
 int x,
 IntBinaryOperator accumulatorFunction)
```

Atomically updates (with memory effects as specified by `VarHandle.compareAndSet(Object...)`) the field of the given object managed by this updater with the results of applying the given function to the current and given values, returning the previous value. The function should be side-effect-free, since it may be re-applied when attempted updates fail due to contention among threads. The function is applied with the current value as its first argument, and the given update as the second argument.

Parameters:

`obj` - An object whose field to get and set

`x` - the update value

`accumulatorFunction` - a side-effect-free function of two arguments

Returns:

the previous value

Since:

1.8

accumulateAndGet

```
public final int accumulateAndGet
(T obj,
 int x,
 IntBinaryOperator accumulatorFunction)
```

Atomically updates (with memory effects as specified by `VarHandle.compareAndSet(Object...)`) the field of the given object managed by this updater with the results of applying the given function to the current and given values, returning the updated value. The function should be side-effect-free, since it may be re-applied when attempted updates fail due to contention among threads. The function is applied with the current value as its first argument, and the given update as the second argument.

Parameters:

`obj` - An object whose field to get and set

`x` - the update value

`accumulatorFunction` - a side-effect-free function of two arguments

Returns:

the updated value

Since:

1.8

Report a bug or suggest an enhancement
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